

Virtual Data Science with Python Apprentice Intern

Week 3 Report

Topic:

Statistical Analysis and Hypothesis Testing in Python

Dataset: Titanic Dataset

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Source: Kaggle

Tools Used: Python, Pandas, SciPy, Matplotlib, Seaborn

Statistical Test: Chi-Square Test of Independence

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1. Introduction

Statistical analysis is an important part of data science because it helps us understand patterns in data and make decisions based on evidence. In this week's task, I performed statistical analysis and hypothesis testing using Python.

For this analysis, I used the Titanic dataset downloaded from the Kaggle website. The dataset contains information about passengers who were on the Titanic, including their passenger ID, survival status, passenger class, name, sex, age, family information, ticket, fare, cabin and embarkation port.

The main purpose of this analysis was to investigate whether passenger gender had a significant relationship with survival. For this purpose, I used the Chi-Square Test of Independence. I also created visualizations using Seaborn and Matplotlib to understand the data more clearly.

The analysis includes the formulation of hypotheses, preparation of the data, application of the Chi-Square test, interpretation of the p-value, and visualization of the results.

2. Dataset Information

The Titanic dataset used in this project was downloaded from **Kaggle**.

The dataset contains **891 passenger records and 12 columns**.

Important columns used in this analysis are:

Column	Description
PassengerId	Unique identification number of a passenger
Survived	Survival status, where 0 = Did not survive and Survived
Pclass	Passenger class: 1, 2 or 3

Sex	Gender of the passenger
Age	Age of the passenger
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Fare	Passenger fare
Cabin	Cabin information
Embarked	Port of embarkation

For the main hypothesis test, I focused mainly on the **Sex** and **Survived** columns because both are categorical variables.

3. Objective

The objective of this analysis is to determine whether there is a statistically significant relationship between passenger gender and survival.

The research question is:

“Was passenger survival significantly associated with gender on the Titanic?”

To answer this question, I used a Chi-Square Test of Independence.

4. Hypothesis

Null Hypothesis (H_0)

There is **no significant relationship** between passenger gender and survival.

In simple words, gender and survival are independent of each other.

Alternative Hypothesis (H_1)

There is a **significant relationship** between passenger gender and survival.

In simple words, survival was associated with passenger gender.

Significance Level

I used a significance level of:

$$\alpha = 0.05$$

Decision rule:

- If **p-value** < **0.05**, reject H_0 .
- If **p-value** $\geq$ **0.05**, do not reject H_0 .

5. Tools and Libraries Used

The following Python libraries were used:

Pandas

Pandas was used for loading and working with the Titanic dataset.

Matplotlib

Matplotlib was used to create and display graphs.

Seaborn

Seaborn was used to create statistical visualizations.

SciPy

SciPy was used to perform the Chi-Square statistical test.

The `chi2_contingency()` function from `scipy.stats` was used for the hypothesis test.

6. Methodology

The analysis was performed using the following steps:

1. Loaded the Titanic dataset using Pandas.
 2. Selected the variables required for the hypothesis.
 3. Created a contingency table between `Sex` and `Survived`.
 4. Applied the Chi-Square Test of Independence.
 5. Calculated the Chi-Square statistic and p-value.
 6. Compared the p-value with the significance level of 0.05.
 7. Created visualizations to support the statistical findings.
 8. Interpreted the results and prepared the final conclusion.
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7. Python Code

Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from scipy.stats import chi2_contingency
```

Loading the Dataset

```
df = pd.read_csv("Titanic-Dataset.csv")
```

The `read_csv()` function is used to read the Titanic CSV file and store it in a Pandas DataFrame named `df`.

8. Creating the Contingency Table

```
table = pd.crosstab(df["Sex"], df["Survived"])

print(table)
```

The `pd.crosstab()` function creates a frequency table showing the number of passengers according to their gender and survival status.

The result was:

Sex	Survived = 0	Survived = 1
Female	81	233
Male	468	109

Here:

- 0 means the passenger did not survive.
- 1 means the passenger survived.

9. Chi-Square Test

The following code was used:

```
chi2, p, dof, expected = chi2_contingency(table)

print("Chi Square Value:", chi2)
print("P-value:", p)
print("Degrees of Freedom:", dof)
print("Expected Frequencies:")
print(expected)
```

The Chi-Square test checks whether two categorical variables are statistically associated.

In this case, the two variables are:

Sex → Gender

Survived → Survival status

10. Chi-Square Test Results

The obtained results were:

Chi-Square Value: 260.71702016732104

P-value: 1.1973570627755645e-58

Degrees of Freedom: 1

The p-value is extremely small and is much less than 0.05.

Therefore:

p-value < 0.05

So, we **reject the Null Hypothesis (H_0)**.

11. Interpretation of the Result

The p-value obtained from the Chi-Square test is approximately:

1.197×10^{-58}

This value is much smaller than the significance level of 0.05.

Therefore, there is strong statistical evidence that passenger gender and survival were associated in the Titanic dataset.

The observed counts also support this result. There were **233 female survivors**, compared with **109 male survivors**. On the other hand, the number of male passengers who did not survive was much higher than the number of female passengers who did not survive.

Therefore, the statistical test and the observed data both support the Alternative Hypothesis.

12. Visualization 1 – Survival by Gender

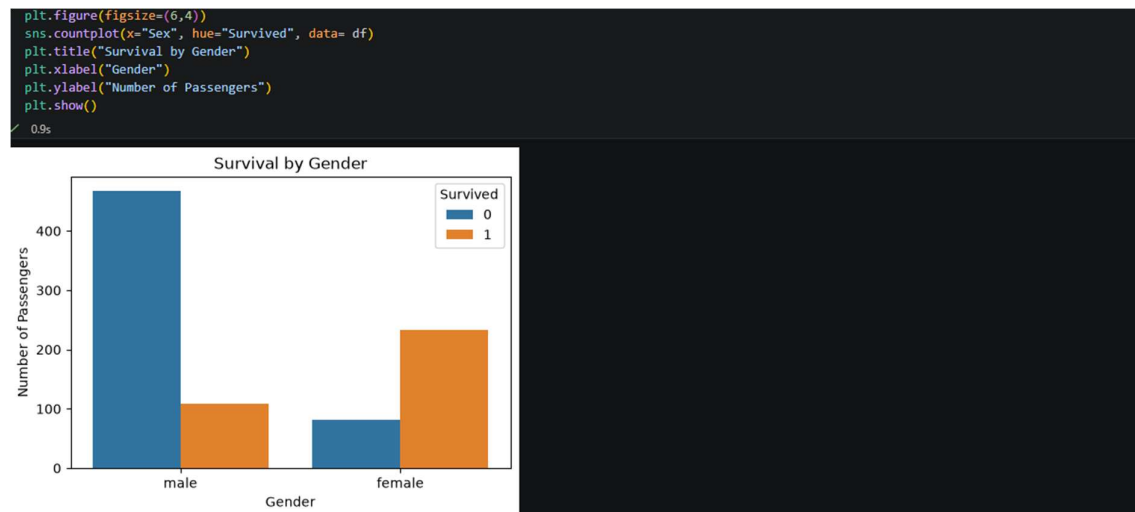
Python Code

```
plt.figure(figsize=(6,4))

sns.countplot(x="Sex", hue="Survived", data=df)

plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```



Interpretation

The graph shows a clear difference between male and female survival.

The number of female survivors is much higher than the number of male survivors. At the same time, the number of male passengers who did not survive is much higher than female passengers who did not survive.

This visualization supports the result obtained from the Chi-Square test and suggests that gender was an important factor associated with survival.

13. Visualization 2 – Survival by Passenger Class

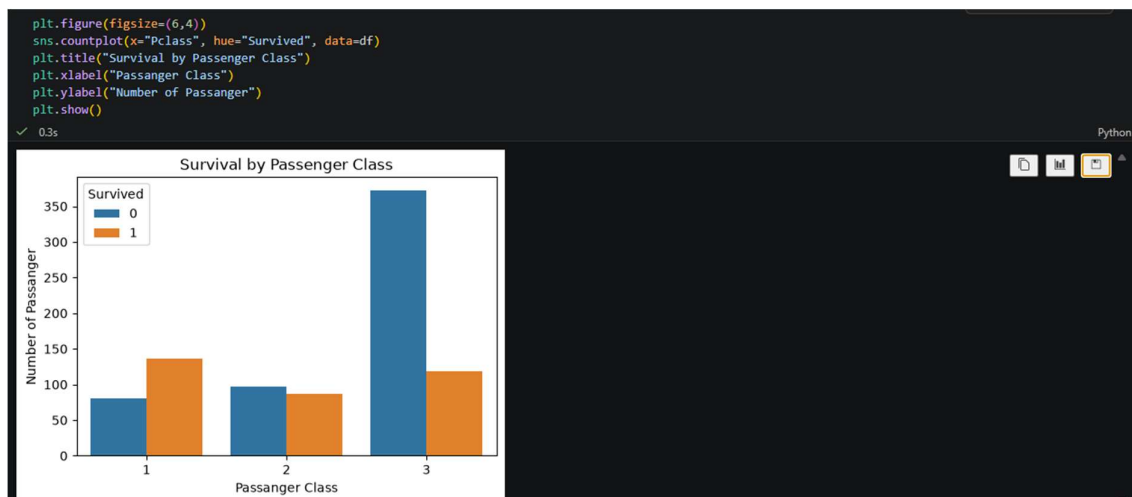
Python Code

```
plt.figure(figsize=(6,4))

sns.countplot(x="Pclass", hue="Survived", data=df)

plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



Interpretation

The graph compares survival outcomes among the three passenger classes.

First-class passengers show a relatively higher number of survivors compared with non-survivors. In contrast, third-class passengers have a much larger number of non-survivors.

This suggests that passenger class may also have influenced survival outcomes. Although passenger class was not the main variable in the hypothesis test, it provides an important area for further analysis.

14. Visualization 3 – Age Distribution

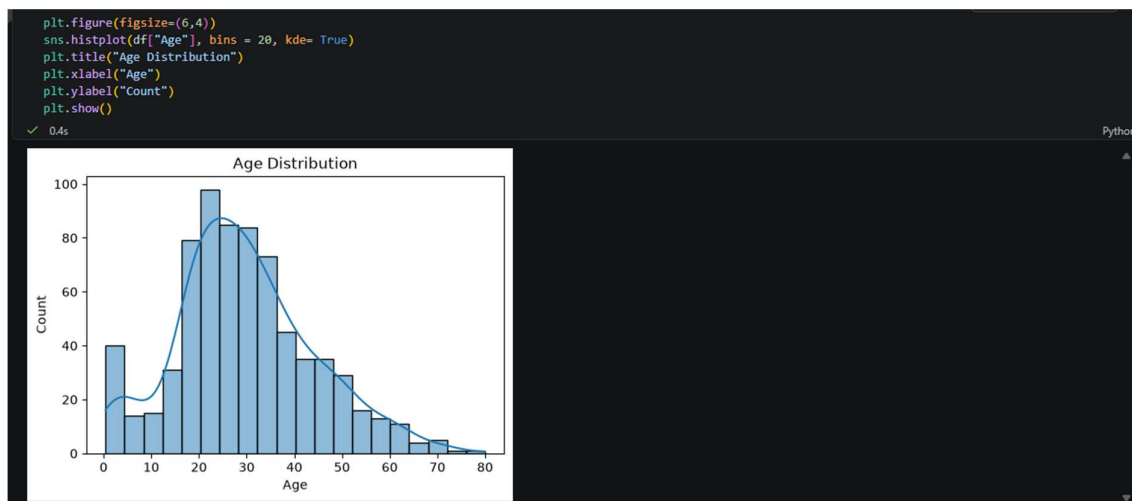
Python Code

```
plt.figure(figsize=(6,4))

sns.histplot(df["Age"], bins=20, kde=True)

plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")

plt.show()
```



Interpretation

The age distribution shows that many passengers were in the younger and middle-age groups.

A large number of observations are concentrated approximately between the teenage years and the thirties. The dataset also contains passengers from a wider range of ages, including older passengers.

This visualization helps in understanding the overall age structure of the passengers and can be useful for further statistical analysis.

15. Overall Findings

From the analysis, the following findings were observed:

1. There is a statistically significant relationship between gender and survival.
2. The Chi-Square test produced a very small p-value.
3. Female passengers had a considerably higher survival count than male passengers.
4. Male passengers had a much higher number of non-survivors.
5. Passenger class also showed noticeable differences in survival outcomes.
6. A large proportion of passengers belonged to the younger and middle-age groups.
7. The visualizations support the statistical findings.

16. Final Conclusion

The main objective of this Week 3 task was to perform statistical analysis and hypothesis testing using Python.

For this analysis, I used the Titanic dataset downloaded from Kaggle and investigated whether passenger gender was significantly associated with survival.

The Chi-Square Test of Independence produced a Chi-Square value of approximately 260.72 and a p-value of approximately 1.20×10^{-58} . Since the p-value is much smaller than the significance level of 0.05, the Null Hypothesis was rejected.

Therefore, the analysis provides strong evidence of a statistically significant relationship between passenger gender and survival in the Titanic dataset.

The visualizations also support this result. Female passengers had a much higher survival count than male passengers. The passenger-class visualization also showed differences in survival outcomes

across classes, while the age distribution provided an overview of the passengers' age groups.

Overall, this task helped me understand how hypothesis testing, p-values, statistical results and visualizations can be combined to draw meaningful conclusions from data.

17. Limitations and Further Analysis

This analysis mainly focused on the relationship between gender and survival. Other variables such as passenger class, age, fare and family information may also have influenced survival.

Further analysis could include:

- Testing the relationship between passenger class and survival.
- Studying the relationship between age and survival.
- Comparing survival rates across different passenger classes.
- Using other statistical tests such as t-tests or ANOVA where appropriate.
- Building a machine learning model to predict passenger survival.

These additional analyses could provide a deeper understanding of the factors that influenced survival.